

WEEK 7 PROJECT REPORT

DATA CLEANING & TRANSFORMATION

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Project Title: Student Performance – Data Cleaning & Transformation

Tool Used: Microsoft Power BI Desktop

Dataset: Healthcare Dataset

INTRODUCTION

The objective of this Week 7 Power BI task was to clean, transform, and prepare the **Student Performance in Exams** dataset for analysis and visualization. The workflow was completed using **Power BI Desktop** and **Power Query Editor**. The dataset was inspected for data quality issues, categorical attributes were standardized, calculated columns were created, student performance was classified, and subject-wise data was reshaped using unpivoting.

IMPORTING THE DATASET

The Student Performance in Exams dataset was imported into **Power BI Desktop** using the **Get Data** option. The appropriate source format, such as Text/CSV or Excel, was selected and the dataset was loaded into Power Query Editor using **Transform Data**.

StudentsPerformance.csv

File Origin

1252: Western European (Windows)

Delimiter

Comma

Data Type Detection

Based on first 200 rows

gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
female	group B	bachelor's degree	standard	none	72	72	74
female	group C	some college	standard	completed	69	90	88
female	group B	master's degree	standard	none	90	95	93
male	group A	associate's degree	free/reduced	none	47	57	44
male	group C	some college	standard	none	76	78	75
female	group B	associate's degree	standard	none	71	83	78
female	group B	some college	standard	completed	88	95	92
male	group B	some college	free/reduced	none	40	43	39
male	group D	high school	free/reduced	completed	64	64	67
female	group B	high school	free/reduced	none	38	60	50
male	group C	associate's degree	standard	none	58	54	52
male	group D	associate's degree	standard	none	40	52	43
female	group B	high school	standard	none	65	81	73
male	group A	some college	standard	completed	78	72	70
female	group A	master's degree	standard	none	50	53	58
female	group C	some high school	standard	none	69	75	78
male	group C	high school	standard	none	88	89	86
female	group B	some high school	free/reduced	none	18	32	28
male	group C	master's degree	free/reduced	completed	46	42	46
female	group C	associate's degree	free/reduced	none	54	58	61

The data in the preview has been truncated due to size limits.

Extract Table Using Examples

Load

Transform Data

Cancel

EXPLORING AND INSPECTING THE DATA

The dataset was examined to understand its structure, columns, rows, and data types. The available attributes included student demographic information and examination scores.

The following types of attributes were identified:

- **Categorical attributes:** Gender, Race/Ethnicity, Parental Education, Lunch, and Test Preparation Course.
- **Numerical attributes:** Math Score, Reading Score, and Writing Score.

The columns were checked for appropriate data types, missing values, invalid values, and duplicate records. This inspection helped identify the transformations required before analysis.

= Table.RenameColumns(#"Reordered Columns",{"Index", "StudentID"}, {"gender", "Gender"}, {"race/ethnicity", "Race/Ethnicity"}, {"parental level of education", "Parental level of Education"}, {"lunch", "Lunch"}, {"test preparation course", "Test Preparation Course"})						
StudentID	Gender	Race/Ethnicity	Parental level of Education	Lunch	Test Preparation Course	
1	female	group B	bachelor's degree	standard	none	
2	female	group C	some college	standard	completed	
3	female	group B	master's degree	standard	none	
4	male	group A	associate's degree	free/reduced	none	
5	male	group C	some college	standard	none	
6	female	group B	associate's degree	standard	none	
7	female	group B	some college	standard	completed	
8	male	group B	some college	free/reduced	none	
9	male	group D	high school	free/reduced	completed	
10	female	group B	high school	free/reduced	none	
11	male	group C	associate's degree	standard	none	
12	male	group D	associate's degree	standard	none	
13	female	group B	high school	standard	none	
14	male	group A	some college	standard	completed	
15	female	group A	master's degree	standard	none	
16	female	group C	some high school	standard	none	
17	male	group C	high school	standard	none	
18	female	group B	some high school	free/reduced	none	
19	male	group C	master's degree	free/reduced	completed	
20	female	group C	associate's degree	free/reduced	none	
21	male	group D	high school	standard	none	
22	female	group B	some college	free/reduced	completed	
23	male	group D	some college	standard	none	
24	female	group C	some high school	standard	none	
25	male	group D	bachelor's degree	free/reduced	completed	

CLEANING THE DATASET

The dataset was cleaned in Power Query to improve its quality and consistency.

The following cleaning operations were performed:

- Duplicate records were removed where appropriate.
- Missing or invalid values were checked and handled.
- Appropriate data types were assigned to the columns.
- Inconsistent categorical values were standardized.
- Column names were reviewed and renamed where necessary for better clarity.

These steps ensured that the dataset was reliable and suitable for further transformation and analysis.

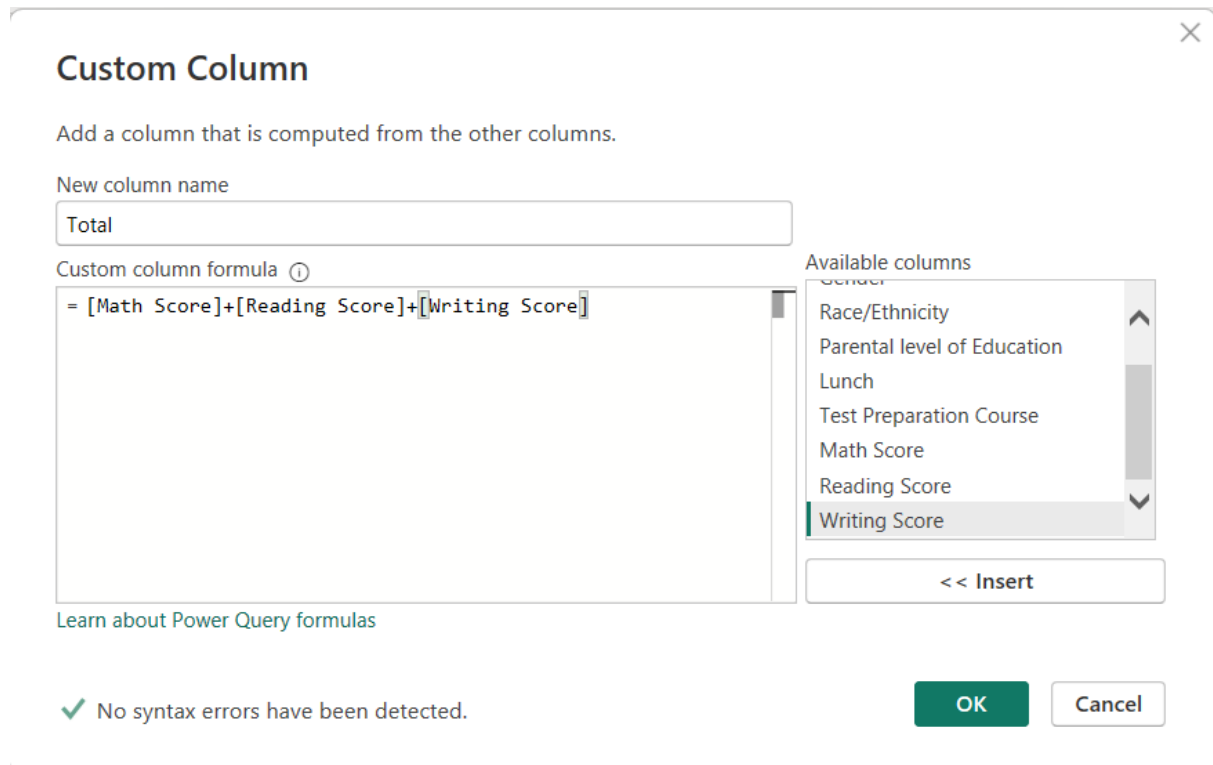
CREATING CALCULATED COLUMNS IN POWER QUERY

1. Creating the Total Score

A Total column was created in Power Query using a Custom Column. The total score was calculated by adding the student's Math, Reading, and Writing scores.

The formula used was: **[Math Score] + [Reading Score] + [Writing Score]**

This provides the combined score of the student across all three subjects.



Custom Column

Add a column that is computed from the other columns.

New column name

Total

Custom column formula ⓘ

= [Math Score]+[Reading Score]+[Writing Score]

Available columns

- Gender
- Race/Ethnicity
- Parental level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

2. Creating the Average Score

An Average column was then created using the Total Score.

The formula used was: **[Total] / 3**

This calculates the student's average performance across Math, Reading, and Writing.

Custom Column

×

Add a column that is computed from the other columns.

New column name

Average

Custom column formula ⓘ

= [Total]/3

Available columns

Parental level of Education

Lunch

Test Preparation Course

Math Score

Reading Score

Writing Score

Total

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK

Cancel

CREATING THE PERFORMANCE GRADE

A Grade column was created using Power Query's Conditional Column feature. The student's Average Score was used to classify performance into different grades.

The conditions used in the completed workflow were:

Average Score	Grade
≥ 90	Grade A
≥ 75	Grade B
≥ 50	Grade C
≥ 40	Grade D
Below 40	Fail

This classification provides a simple way to identify the performance level of each student.

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Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value	Output
If	Average	is greater than or...	90	Grade A
Else If	Average	is greater than or...	75	Grade B
Else If	Average	is greater than or...	50	Grade C
Else If	Average	is greater than or...	40	Grade D

Else

CREATING PASS/FAIL CLASSIFICATION

A Pass/Fail column was created using a Conditional Column.

The criterion used was:

- Average > 40 → Pass
- Otherwise → Fail

This classification allows students to be easily separated into passing and failing groups.

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Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

	Column Name	Operator	Value	Output
If	Average	is greater than	40	Pass

Else

RESHAPING SUBJECT DATA USING UNPIVOT

To make the dataset more suitable for subject-wise analysis, the three subject score columns were transformed using the **Unpivot Columns** operation.

The following columns were selected:

- Math Score
- Reading Score
- Writing Score

After unpivoting, the resulting columns were renamed appropriately:

- **Subject**
- **Score**

This transformation converted the subject columns into rows, allowing the performance of students to be analyzed consistently across different subjects.

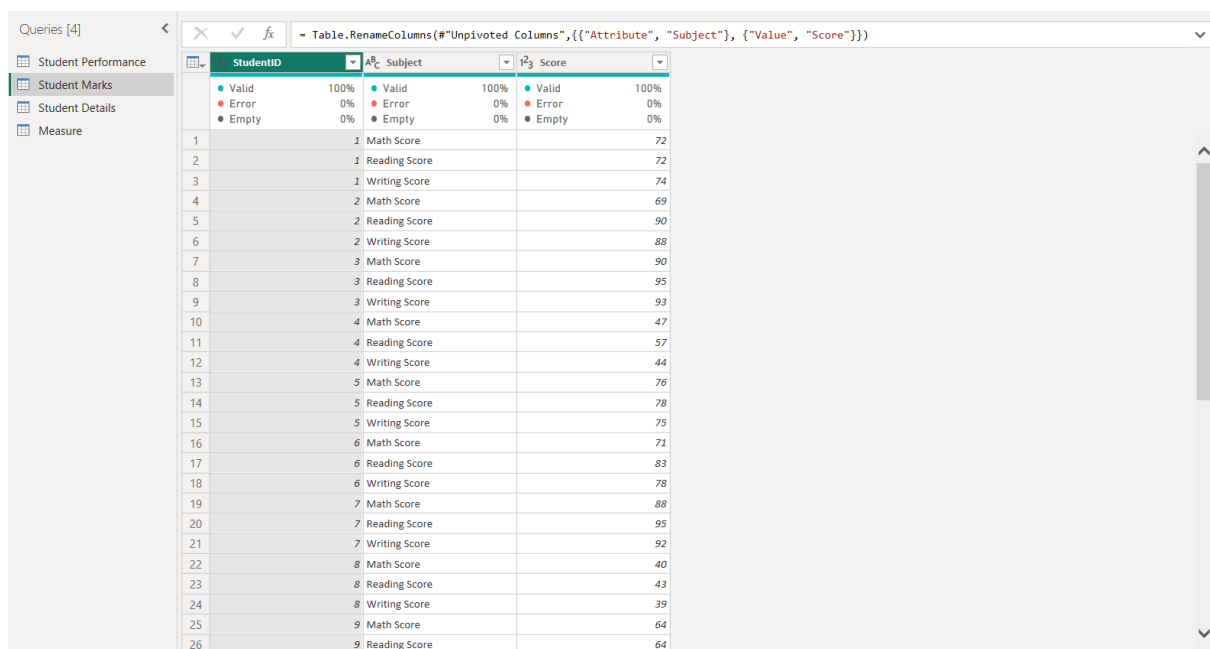


Table: Table.RenameColumns(#Unpivoted Columns,{{"Attribute", "Subject"}, {"Value", "Score"}})

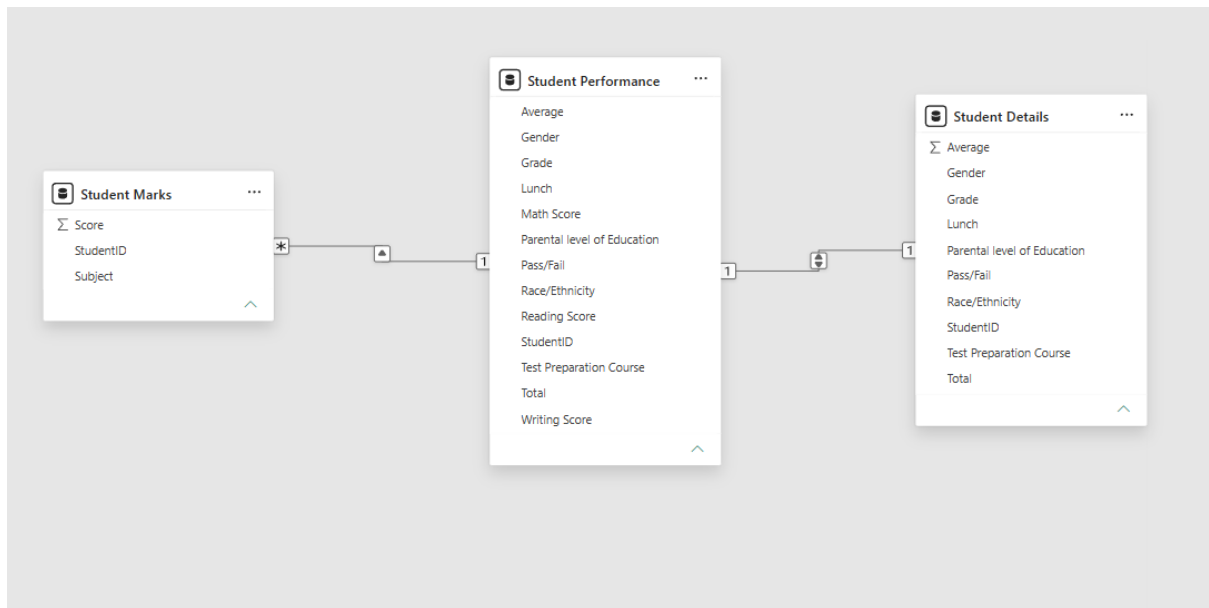
StudentID	Subject	Score
1	Math Score	72
2	Reading Score	72
3	Writing Score	74
4	Math Score	69
5	Reading Score	90
6	Writing Score	88
7	Math Score	90
8	Reading Score	95
9	Writing Score	93
10	Math Score	47
11	Reading Score	57
12	Writing Score	44
13	Math Score	76
14	Reading Score	78
15	Writing Score	75
16	Math Score	71
17	Reading Score	83
18	Writing Score	78
19	Math Score	88
20	Reading Score	95
21	Writing Score	92
22	Math Score	40
23	Reading Score	43
24	Writing Score	39
25	Math Score	64
26	Reading Score	64

PREPARING THE DATA MODEL

After completing the transformations, the resulting table was reviewed to ensure that the structure was appropriate for analysis.

Unnecessary columns were removed where required, and the transformed fields were checked for appropriate data types and consistency. The resulting structure was prepared for use within the Power BI data model.

The model was reviewed to ensure that the required fields were available for calculations, filtering, and visualization.



CREATING DAX MEASURES

Basic DAX measures were created to calculate important performance indicators.

1. Average Math Score

This measure calculates the average score obtained by students in Mathematics.

2. Average Reading Score

This measure calculates the average score obtained by students in Reading.

3. Average Writing Score

This measure calculates the average score obtained by students in Writing.

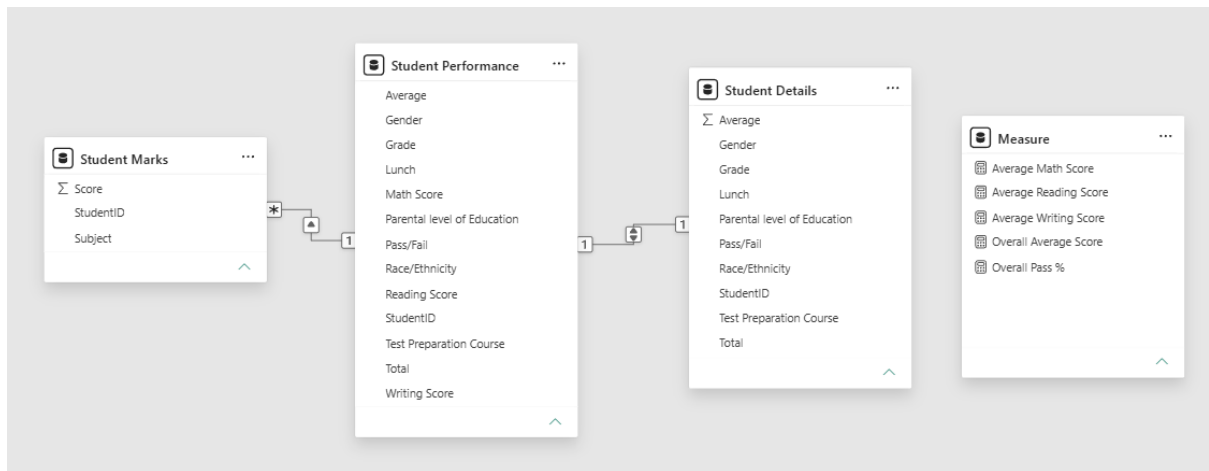
4. Overall Average Score

This measure represents the overall average performance of students.

5. Overall Pass %

This measure calculates the percentage of students who passed according to the defined pass criterion.

These measures provide key indicators that can later be used in Power BI cards, charts, and other visualizations.



CONCLUSION

The Week 7 task successfully demonstrated the process of preparing a real-world student performance dataset for analysis using Power BI. Data cleaning techniques were applied to improve data quality, categorical fields were standardized, calculated columns were created, students were classified based on performance and pass/fail status, and subject data was reshaped using unpivoting.

The creation of DAX measures further enhanced the analytical capabilities of the dataset. Overall, the completed workflow produced a structured and analysis-ready dataset that can be effectively used for student performance visualization and dashboard development.